MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6688

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, miniature, RF, sharp cutoff

Outline	---	6-1 (EIA)
Base	---	E9-1
Envelope	---	T6-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	k	g1	k	h	h	int con	a	g3	g2

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Ehk	ec1
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	v
Maximum:	6.6	210	0	175	0	60	---
Minimum:	6.0	---	-50	---	---	---	-100
TEST CONDITIONS:	6.3	190	+9	160	0	0	---

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Rk	Rg1	Ik	Pp	Pg2	TE	Alt
Unit:	Ohms	Meg	mAdc	W	W	°C	ft
Maximum:	---	0.1	25	2.7	0.6	+155	(see note 1)
Minimum:	---	---	---	---	---	---	---
TEST CONDITIONS:	630	---	---	---	---	---	---

GENERAL:

Qualification - Required

6688

FSC 5960

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1301	Heater current		0.4	II	If	285	315	mA
1336	Heater-cathode leakage		0.4	II	Ihk	---	15	µAdc
1266	Total grid current	See note 2	0.4	II	Ic1	0	-0.5	µAdc
1256	Anode current (1)	Ebb = 180 Vdc; Ecc1 = 0; Ecc2 = 150 Vdc; Ecc3 = 0; Rk = 78 Ohms See note 2	0.4	II	Ib	8.0	18.0	mAdc
1256	Anode current (2)	Ec1 = -6.0 Vdc; Ebb = 180 Vdc; Ecc2 = 150 Vdc; Ecc3 = 0; Rk = 78 ohms	0.4	II	Ib	---	100	µAdc
1256	Screen-grid current		0.4	II	Ic2	2.9	3.7	mAdc
1306	Transconductance (1)		0.4	II	Sm	14200	18800	µmhos
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes		2.5	I	R	20	---	Meg
1306	Transconductance (2)	Ef = 5.7 V	2.5	I	ΔSm Ef	---	15	%
1266	Grid emission	Ef = 7.5 V; Ec1 = -6.0 Vdc; Rk = 0 (see note 3)	2.5	I	Ic1	0	-0.5	µAdc
2201	Noise and microphonics	Ef = 6.3 V; Ehk = 0; Ebb = Ecc2 = 200 Vdc; Ec1 = 0; Eca1 = 600 mVac; Rk = 1,000 ohms; Rp = 0.1 Meg; Rg2 = 0.5 Meg; Cg2 = 2 µf (see note 4)	2.5	I	---	---	---	---
1331	Direct-interelectrode capacitance	Shield No. 315 (see note 5)	6.5	Code E	Cg1p Cin Cout	--- 6.7 2.50	0.030 8.5 3.50	pF pF pF
1031	Low-frequency vibration	Rp = 2,000 ohms	6.5	Code E	Ep	---	300	mVdc
1041	Shock	450 G; Ehk = +100 Vdc (see note 7)	---	See note 6	---	---	---	---
1031	Vibration fatigue	2.5 G; F = 25 min, 60 max; fixed frequency	6.5	See note 6	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance Inspection, part 2 -Continued							
---	Post-shock and vibration fatigue end points							
1031	Low-frequency vibration		---	---	Ep	---	400	mVac
1336	Heater-cathode leakage		---	---	Ihk	---	30	µAdc
1306	Transconductance (1)		---	---	Sm	12500	---	µmhos
1266	Total grid current		---	---	Ic1	0	-1.0	µAdc
1121	Base strain		---	---	---	---	---	---
2126	Glass strain		6.5	I	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance Inspection, part 3					
1506	Heater-cycling life	Ef = 7.5 V; Ehk = 60 Vdc; Eb = Ec1 = Ec2 = 0	---	---	---	---
---	Life-test end point (heater-cycling)					
1336	Heater-cathode leakage		Ihk	---	30	µAdc
1516	Stability life	Eb = 200 Vdc; Ec1 = +9 Vdc; Ec2 = 170 Vdc; Ec3 = 0; Rk = 680 ohms; Rg1 = 0.5 Meg; TA = room	---	---	---	---
---	Life-test end point (stability)					
1306	Change in transconductance (1) of individual tubes		ΔSm t	---	10	%
1501	Intermittent life	Stability life-test conditions; TE = +155°C (min) (see notes 8 and 9)	---	---	---	---
---	Life-test end points (intermittent) (500 hours)					
---	Inoperatives		---	---	---	---
1266	Total grid current		Ic1	0	-0.9	µAdc
1301	Heater current		If	285	315	mA
1306	Change in transconductance (1) of individual tubes		ΔSm t	---	20	%
1306	Transconductance (2)		ΔSm Ff	---	15	%
1336	Heater-cathode leakage		Ihk	---	30	µAdc
1211	Insulation of electrodes		R	10	---	Meg
1306	Transconductance (1) average change		Avg ΔSm t	---	15	%
---	Total defectives		---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
	Quality conformance inspection, part 3 - Continued					
---	Life-test end points (intermittent) (1,000 hours)					
---	Inoperatives		---	---	---	---
1266	Total grid current		Ic1	0	-0.9	µAdc
1301	Heater current		If	285	315	mA
1306	Change in transconductance (1) of individual tubes		ΔSm t	---	25	%
1336	Heater-cathode leakage		Ihk	---	30	µAdc
1211	Insulation of electrodes		R	10	---	Meg
---	Total defectives		---	---	---	---

NOTES:

1. See "Reduced pressure (altitude) rating," and altitude, maximum peak voltage.
2. This test shall be performed at the conclusion of the holding period.
3. Prior to this test, tubes shall be preheated a minimum of 5 minutes at the conditions indicated below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef Vdc	Ec1 Vdc	Ec2 Vdc	Eb Vdc	Rk Ohms
7.5	+9	160	190	630

4. The rejection level shall be set at the VU meter reading obtained during calibration.
5. Pin 6 floating.
6. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12-months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 12-month rule shall apply. MIL-STD-105, sample size code letter E, shall apply.
7. A grid resistor of 0.1 megohm shall be added; however, this resistor will not be used when a thyratron-type short indicator is employed.
8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib (+5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.
9. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, due to failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed three.
10. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Air Force - 85

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(Project 5960-3221)

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